Installation d'un Firewall pfSense

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BTS2 SISR

Objectifs :

Accéder à votre session Proxmov VE

Installer pfSense sur une machine virtuelle

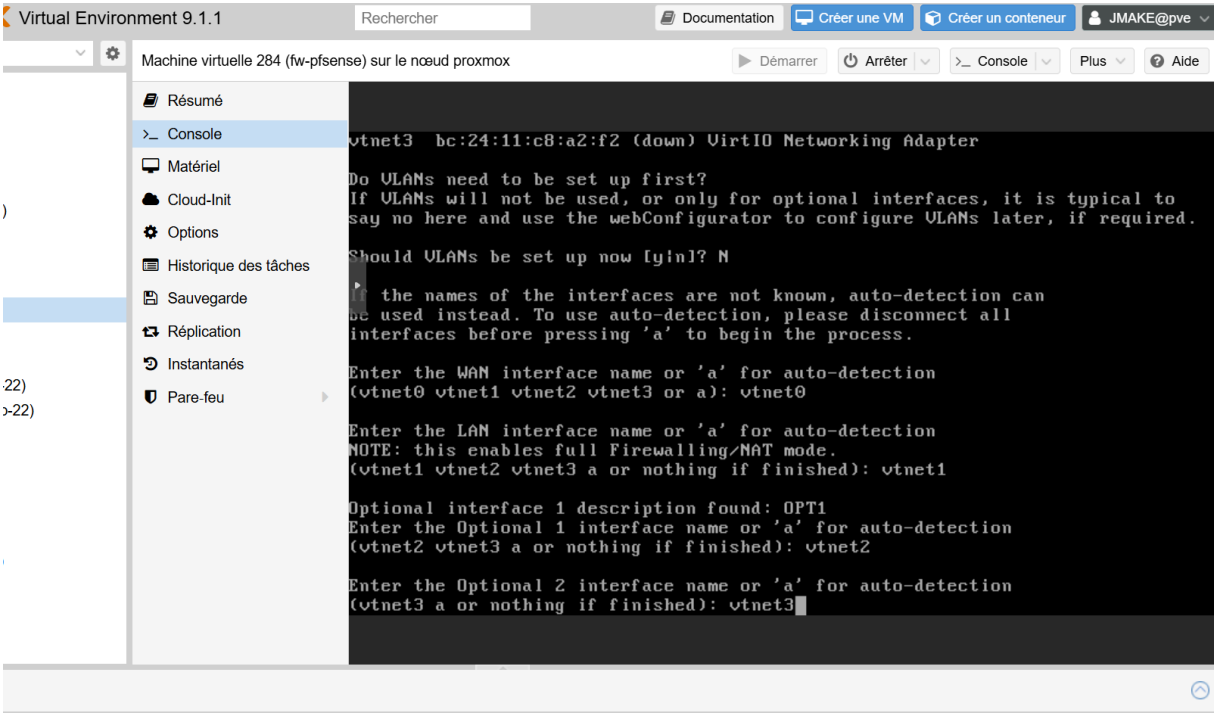
Configure le NAT

Configurer une DMZ

Le filtrage Inter-VLAN

Tester la sécurité réseau

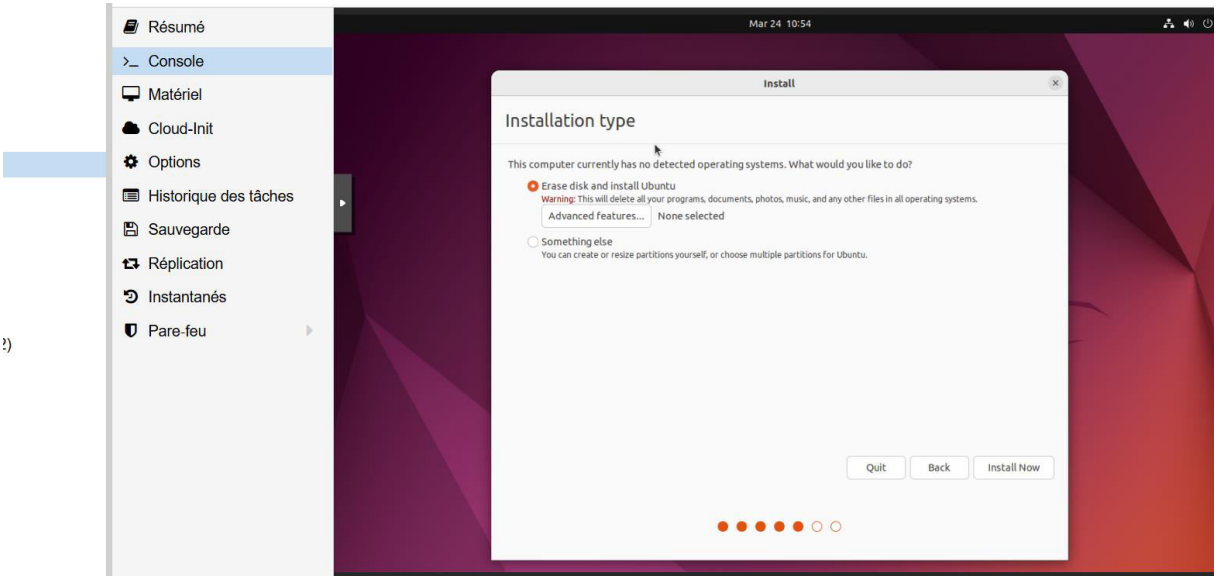
1. Création des VLANs sur Proxmox :



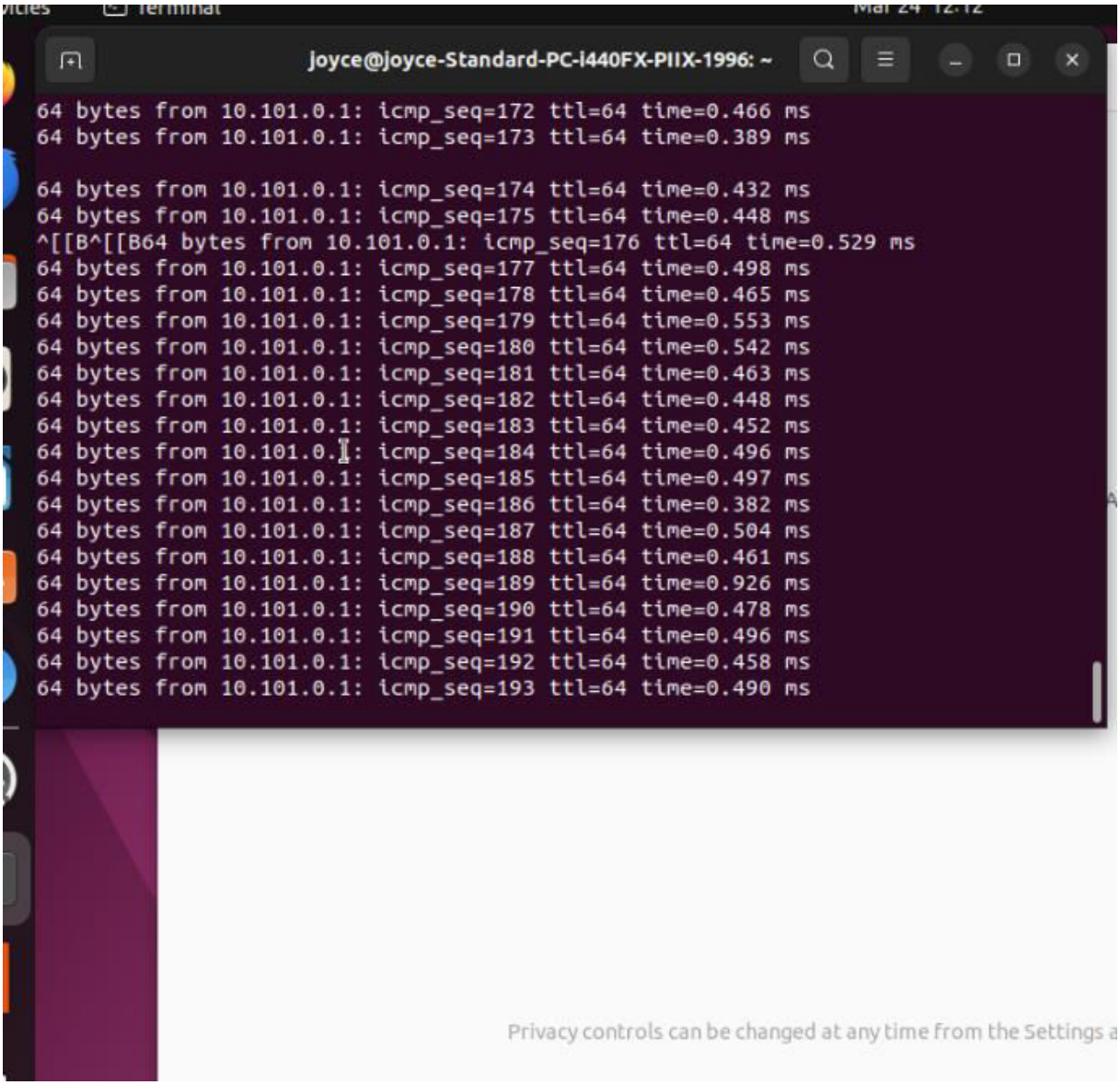
Assignation des interfaces

```
*** Welcome to pfSense 2.6.0-RELEASE (amd64) on pfSense ***

WAN (wan)       -> vtnet0     -> v4/DHCP4: 172.16.1.22/16
LAN (lan)       -> vtnet1     -> v4: 10.101.0.1/24
OPT1 (opt1)     -> vtnet2     -> v4: 10.102.0.1/24
OPT2 (opt2)     -> vtnet3     -> v4: 10.103.0.1/24
```

Adressage adresses IP

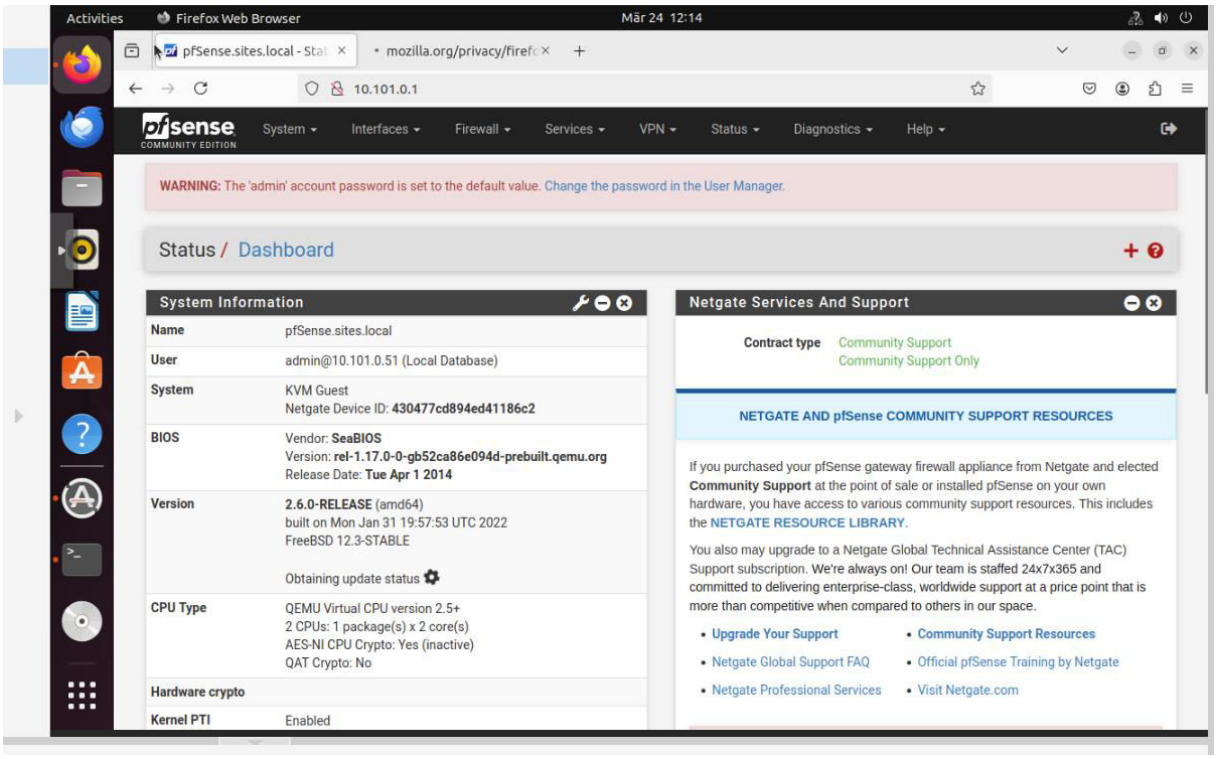


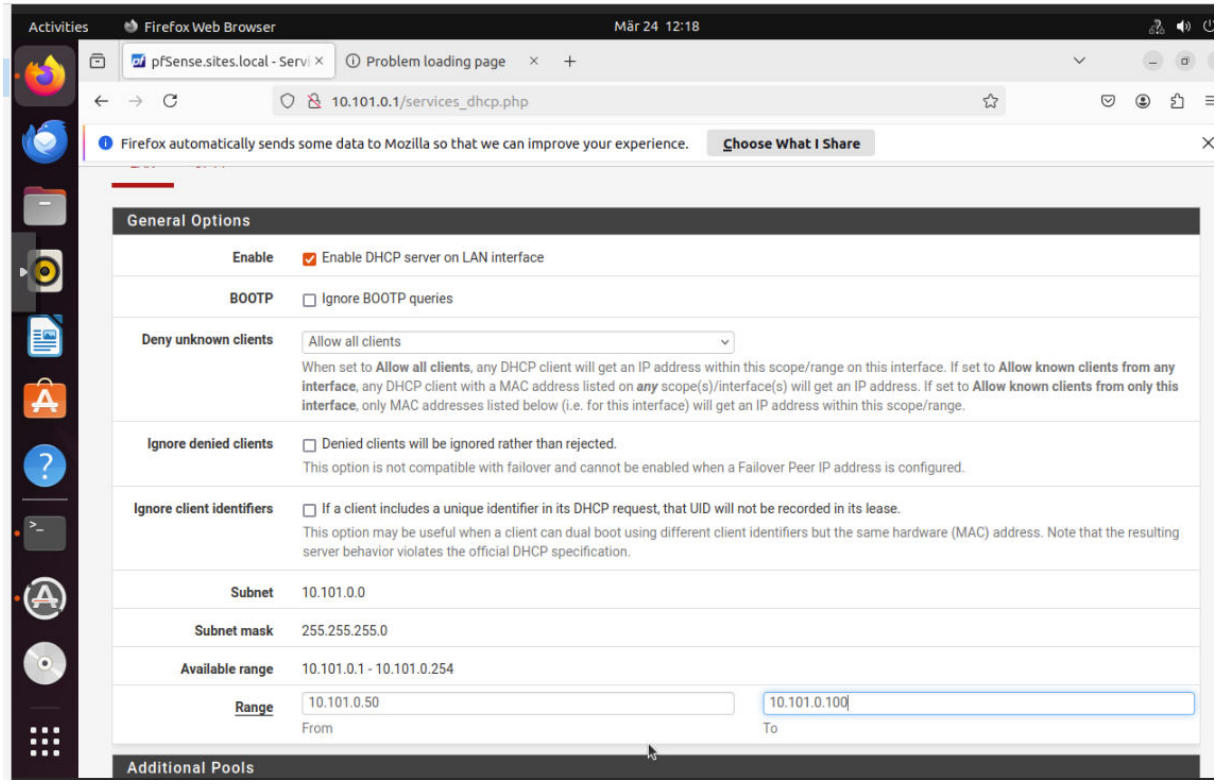
Installation Ubuntu



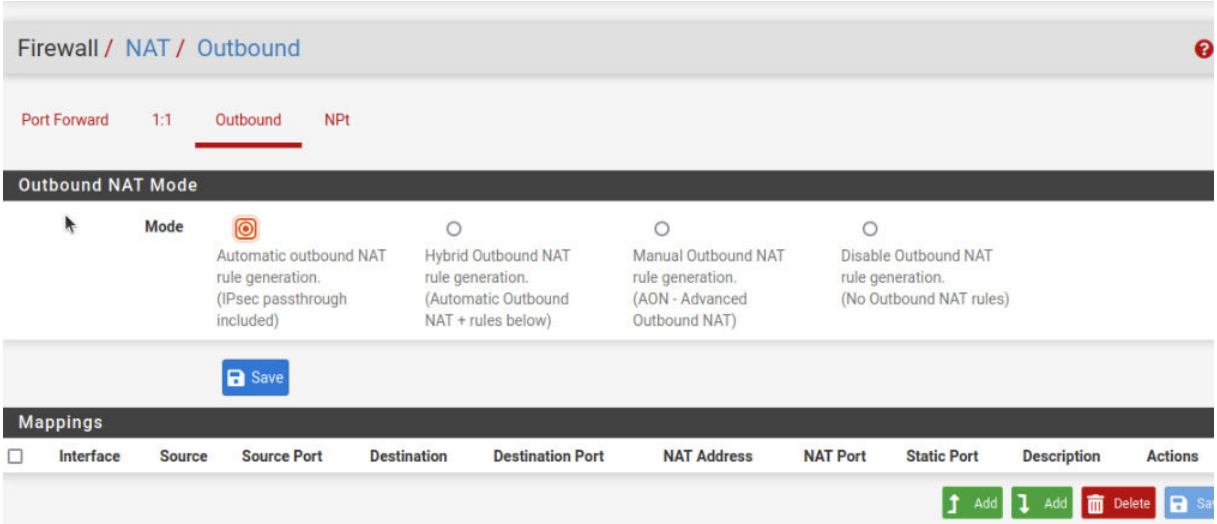
Vérification sur le PC-LAN

2. **<u>Accès à pfSense depuis le PC-LAN :</u>**

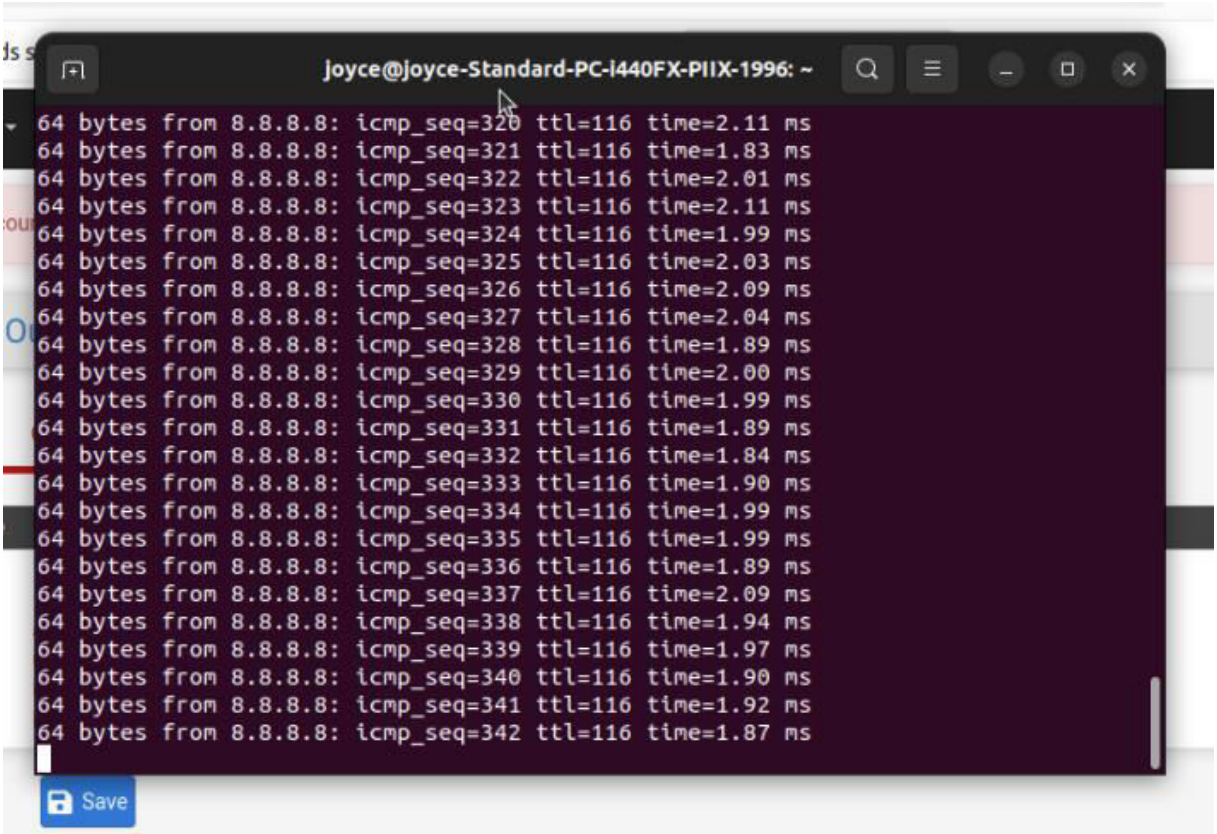

Activities
Firefox Web Browser
Mär 24 12:14
pfSense.sites.local - Stat
mozilla.org/privacy/firefo
10.101.0.1
pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.
Status / Dashboard
System Information
Name pfSense.sites.local
User admin@10.101.0.51 (Local Database)
System KVM Guest
Netgate Device ID: 430477cd894ed41186c2
BIOS Vendor: SeaBIOS
Version: rel-1.17.0-0-gb52ca86e094d-prebuilt.qemu.org
Release Date: Tue Apr 1 2014
Version 2.6.0-RELEASE (amd64)
built on Mon Jan 31 19:57:53 UTC 2022
FreeBSD 12.3-STABLE
Obtaining update status
CPU Type QEMU Virtual CPU version 2.5+
2 CPUs: 1 package(s) x 2 core(s)
AES-NI CPU Crypto: Yes (inactive)
QAT Crypto: No
Hardware crypto
Kernel PTI Enabled
Netgate Services And Support
Contract type Community Support
Community Support Only
NETGATE AND pfSense COMMUNITY SUPPORT RESOURCES
If you purchased your pfSense gateway firewall appliance from Netgate and elected Community Support at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the NETGATE RESOURCE LIBRARY.
You also may upgrade to a Netgate Global Technical Assistance Center (TAC) Support subscription. We're always on! Our team is staffed 24x7x365 and committed to delivering enterprise-class, worldwide support at a price point that is more than competitive when compared to others in our space.
Upgrade Your Support
Community Support Resources
Netgate Global Support FAQ
Official pfSense Training by Netgate
Netgate Professional Services
Visit Netgate.com


Activities
Firefox Web Browser
Mär 24 12:18
pfSense.sites.local - Servi
Problem loading page
10.101.0.1/services_dhcp.php
Firefox automatically sends some data to Mozilla so that we can improve your experience.
Choose What I Share
General Options
Enable
Enable DHCP server on LAN interface
BOOTP
Ignore BOOTP queries
Deny unknown clients
Allow all clients
When set to Allow all clients, any DHCP client will get an IP address within this scope/range on this interface. If set to Allow known clients from any interface, any DHCP client with a MAC address listed on any scope(s)/interface(s) will get an IP address. If set to Allow known clients from only this interface, only MAC addresses listed below (i.e. for this interface) will get an IP address within this scope/range.
Ignore denied clients
Denied clients will be ignored rather than rejected.
This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.
Ignore client identifiers
If a client includes a unique identifier in its DHCP request, that UID will not be recorded in its lease.
This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.
Subnet 10.101.0.0
Subnet mask 255.255.255.0
Available range 10.101.0.1 - 10.101.0.254
Range 10.101.0.50 From
10.101.0.100 To
Additional Pools

DHCP sur LAN

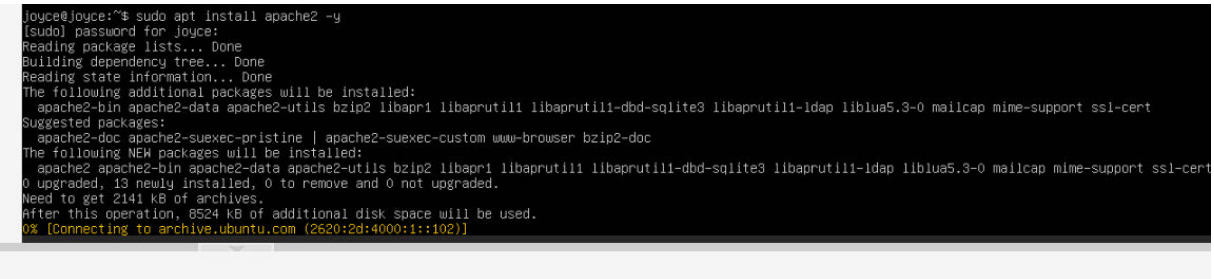


NAT sortant

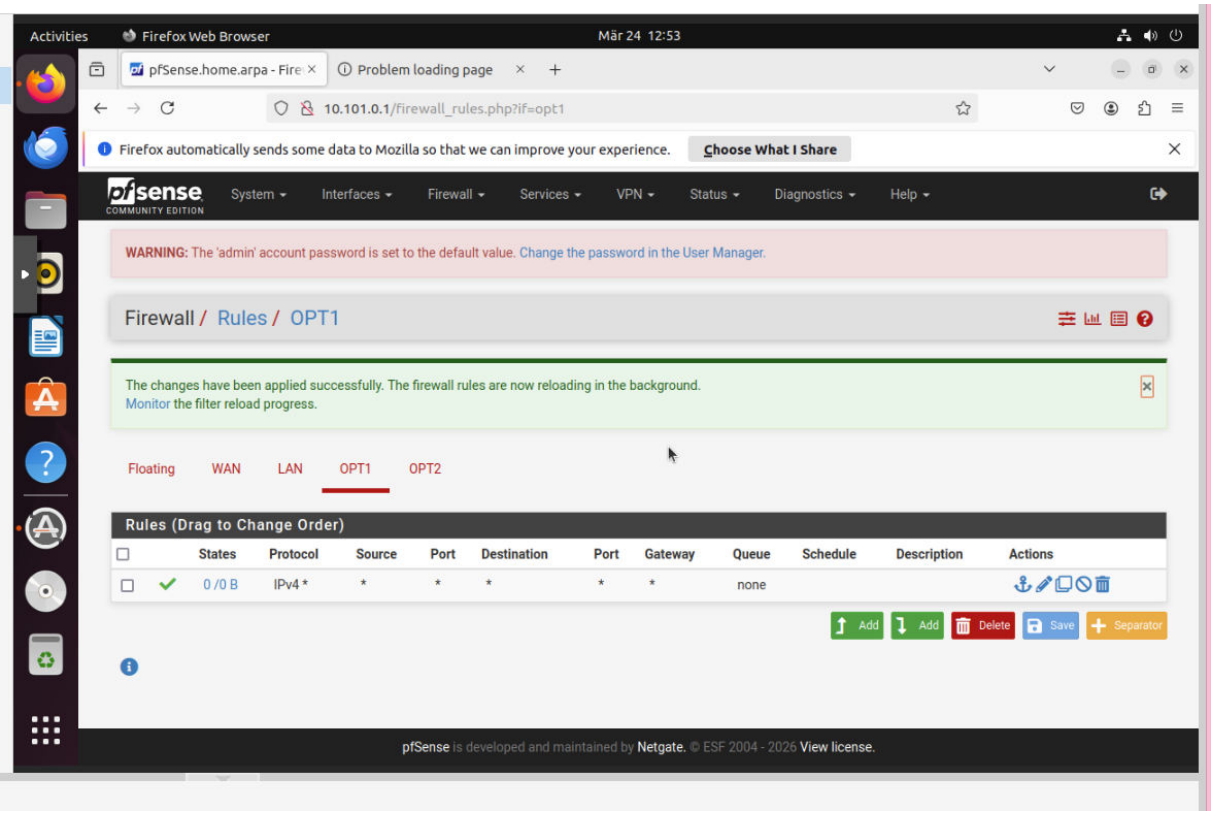


Ping 8.8.8.8

Mise en place de la DMZ et règles firewall inter-VLANs :



Installation serveur web apache

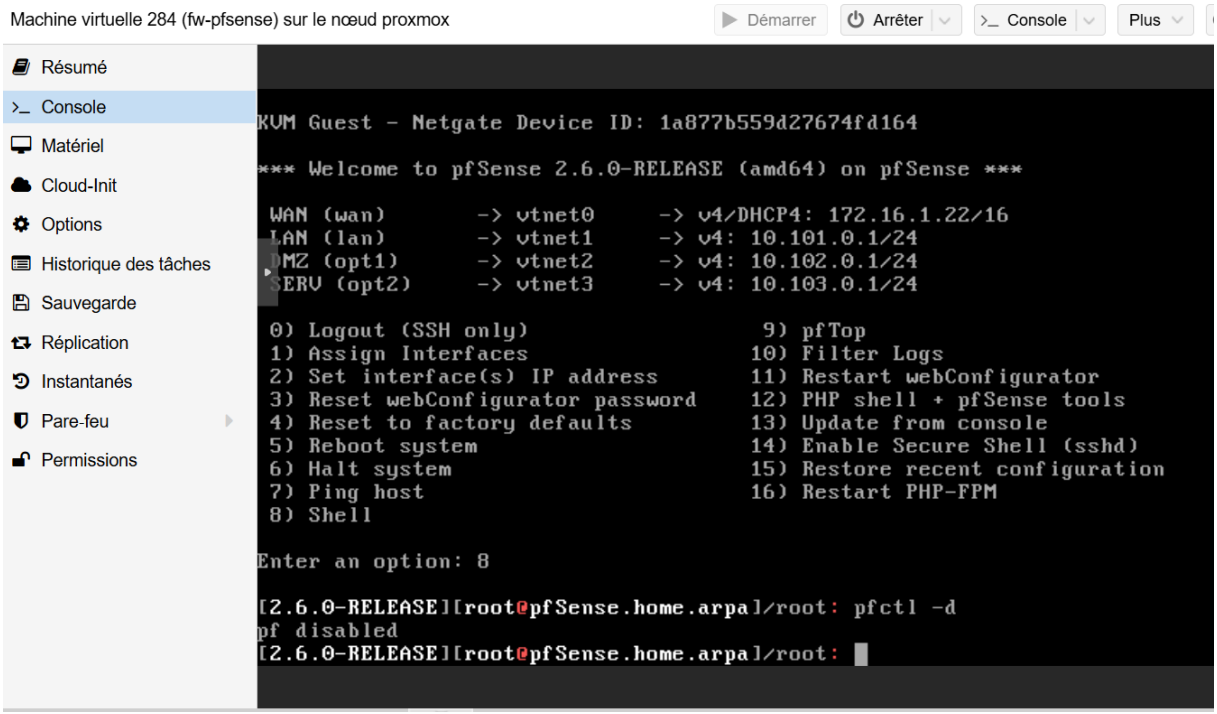


Mise en place d’une règle pour autoriser l’accès à Internet pour pouvoir installer web apache

```
Setting up libaprutil1:amd64 (1.6.1-5ubuntu4.22.04.2) ...
Setting up mime-support (3.66) ...
Setting up libaprutil1-ldap:amd64 (1.6.1-5ubuntu4.22.04.2) ...
Setting up libaprutil1-dbd-sqlite3:amd64 (1.6.1-5ubuntu4.22.04.2) ...
Setting up apache2-utils (2.4.52-1ubuntu4.19) ...
Setting up apache2-bin (2.4.52-1ubuntu4.19) ...
Setting up apache2 (2.4.52-1ubuntu4.19) ...
Enabling module mpm_event.
Enabling module authz_core.
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
joyce@joyce:~$ _
```

Installation réussie

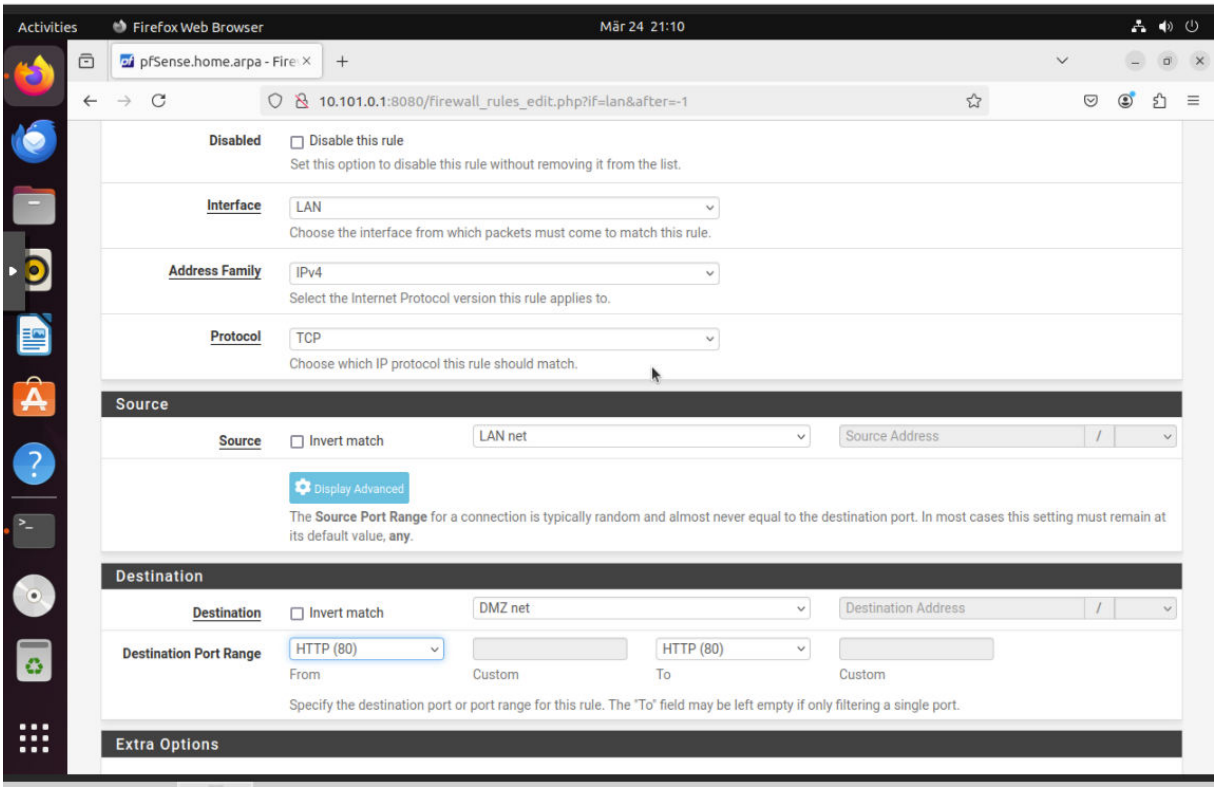
Conflit rencontré, conflit de flux suite à l'installation d'Apache sur mon serveur DMZ.



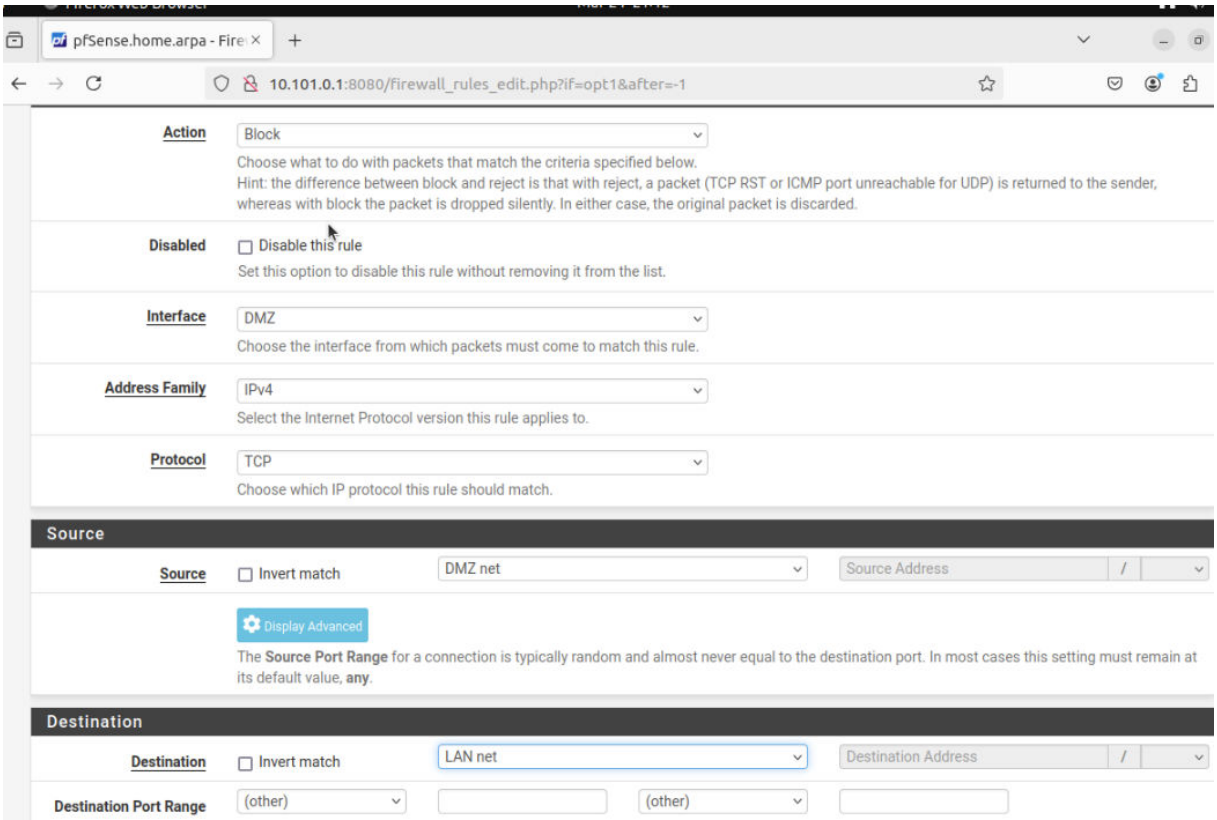
Remédiation par une désactivation temporaire du firewall

```
joyce@joyce-Standard-PC-i440FX-PIIX-1996:~$ sudo dhclient -v ens18
Internet Systems Consortium DHCP Client 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/ens18/bc:24:11:ae:32:8c
Sending on   LPF/ens18/bc:24:11:ae:32:8c
Sending on   Socket/fallback
DHCPREQUEST for 10.101.0.104 on ens18 to 255.255.255.255 port 67 (xid=0x5035536f
)
DHCPACK of 10.101.0.104 from 10.101.0.1 (xid=0x6f533550)
Error: ipv4: Address already assigned.
```

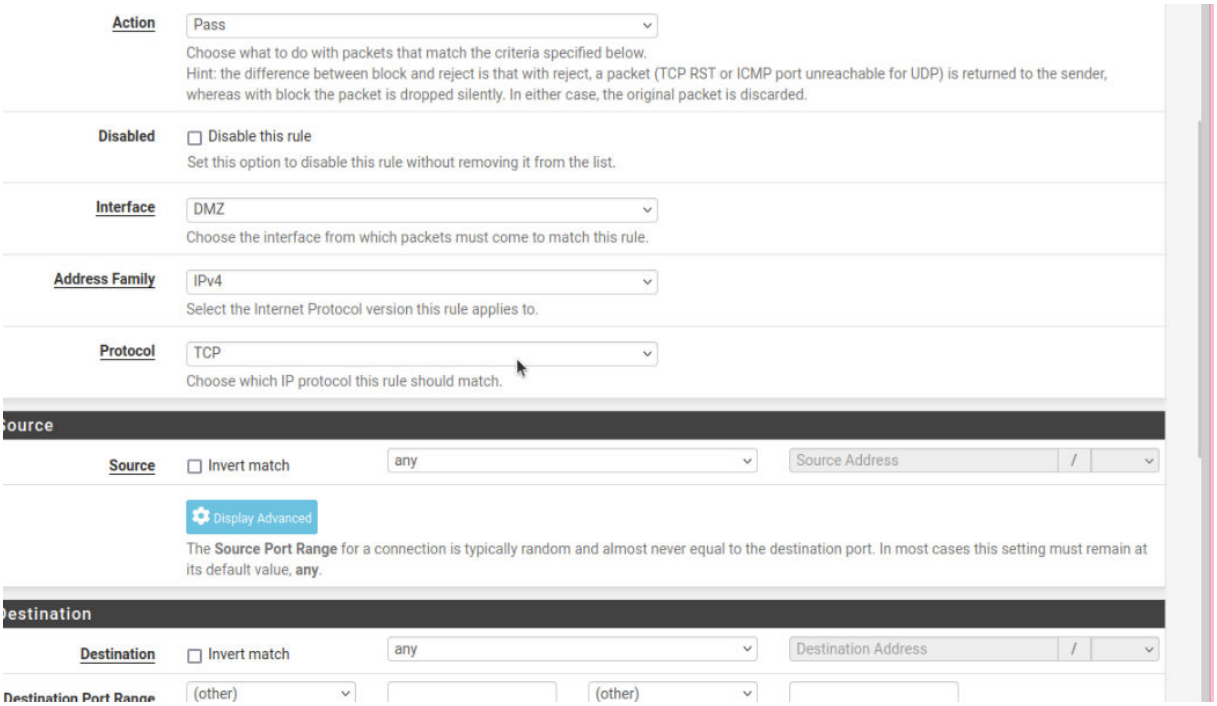
Adresse IP à nouveau à jour



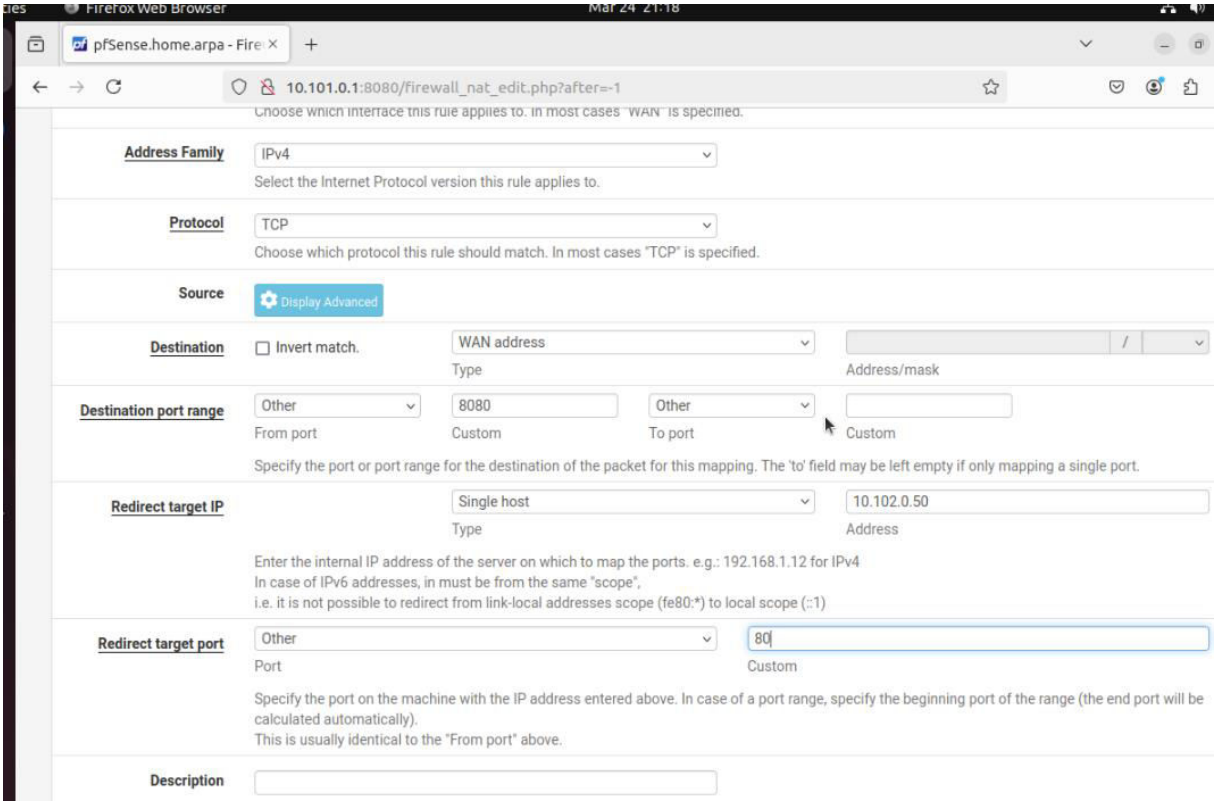
Autoriser la connexion à partir du : LAN → vers DMZ en HTTP uniquement.



Bloquer la connexion depuis la DMZ → LAN



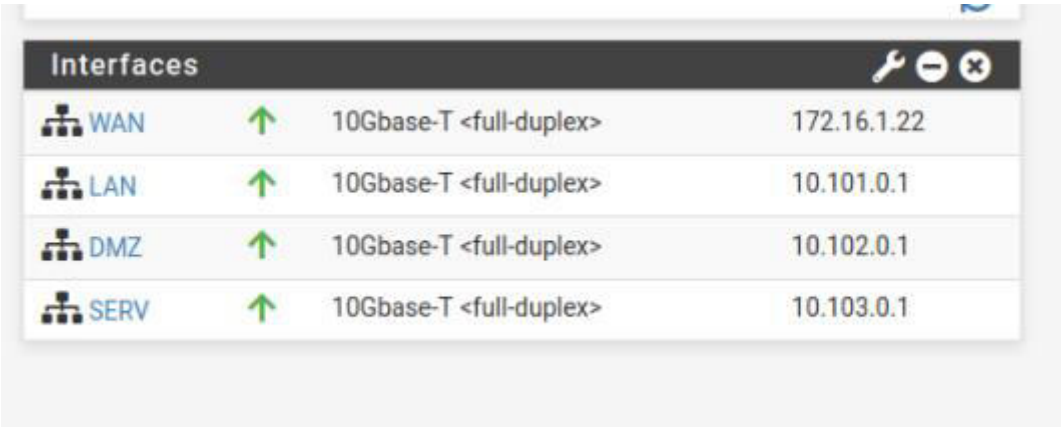
Autoriser la connexion depuis : DMZ → Internet



Nat entrant (Internet → DMZ) :

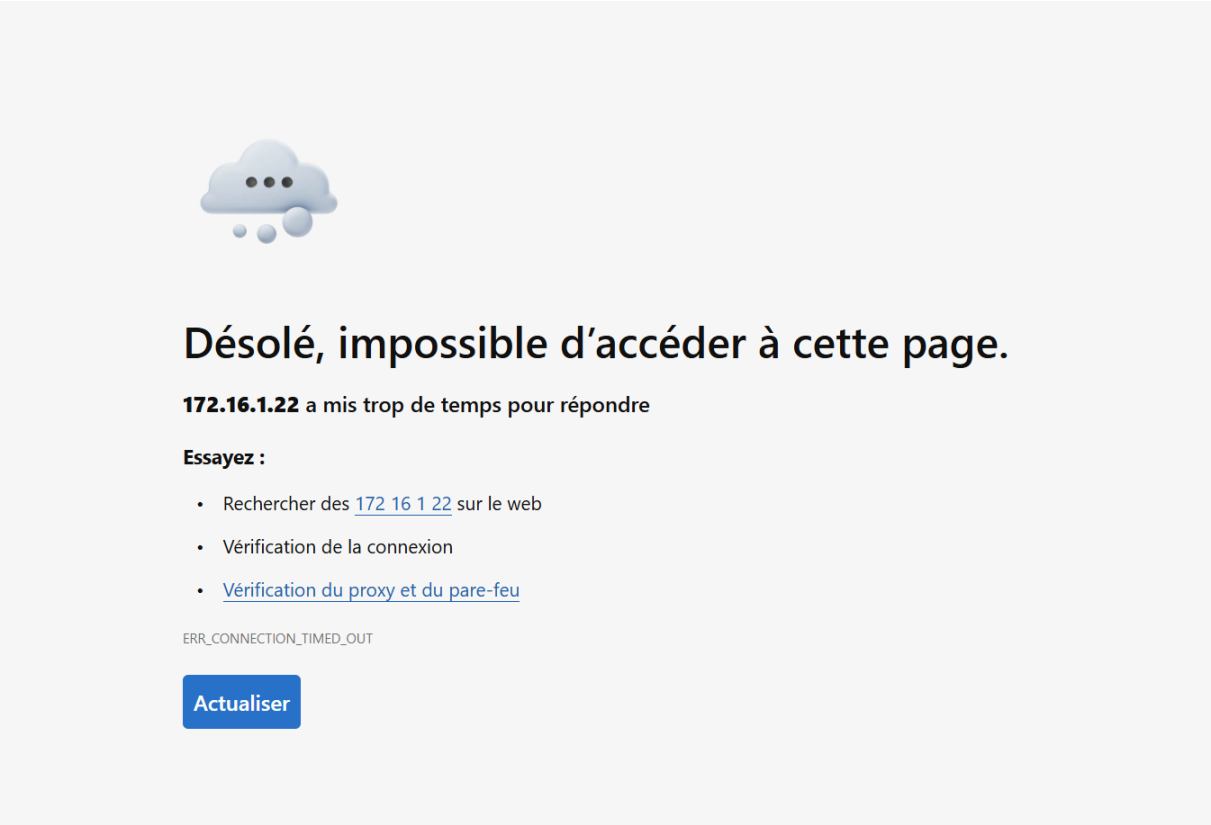
http://172.16.1.22:8080

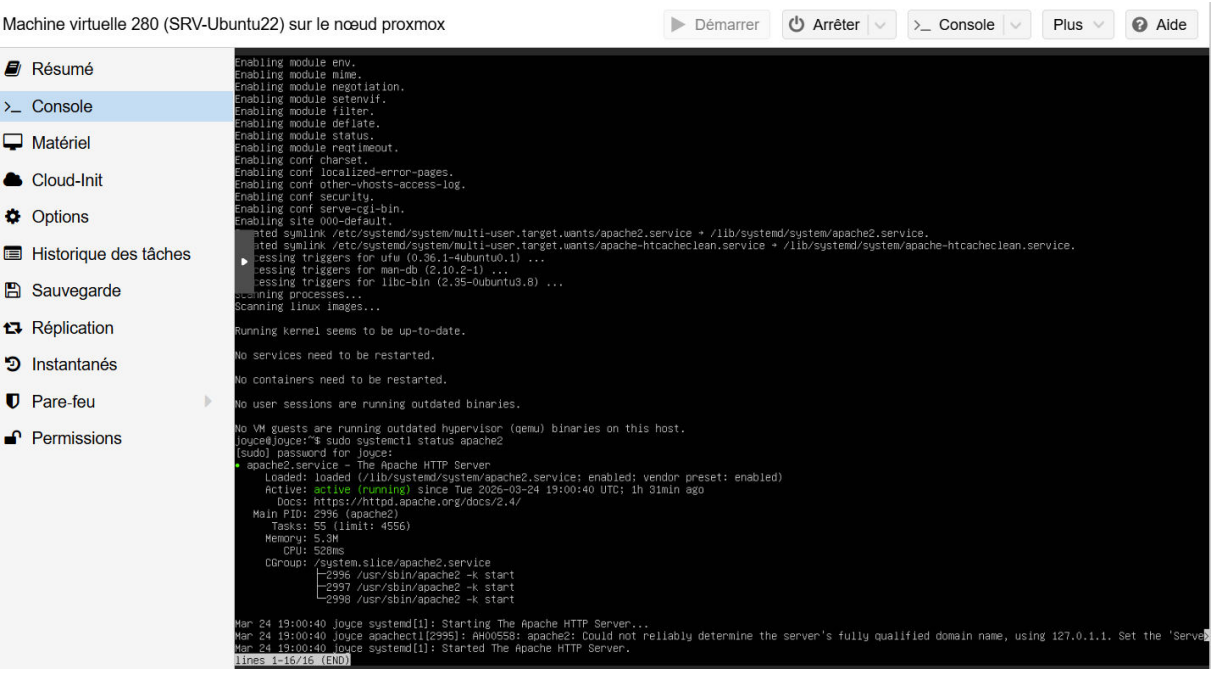
Tests finaux



Interfaces			
WAN	↑	10Gbase-T <full-duplex>	172.16.1.22
LAN	↑	10Gbase-T <full-duplex>	10.101.0.1
DMZ	↑	10Gbase-T <full-duplex>	10.102.0.1
SERV	↑	10Gbase-T <full-duplex>	10.103.0.1

Le lien n'étant pas responsif, je vais vérifier le serveur ainsi que les pare feux.

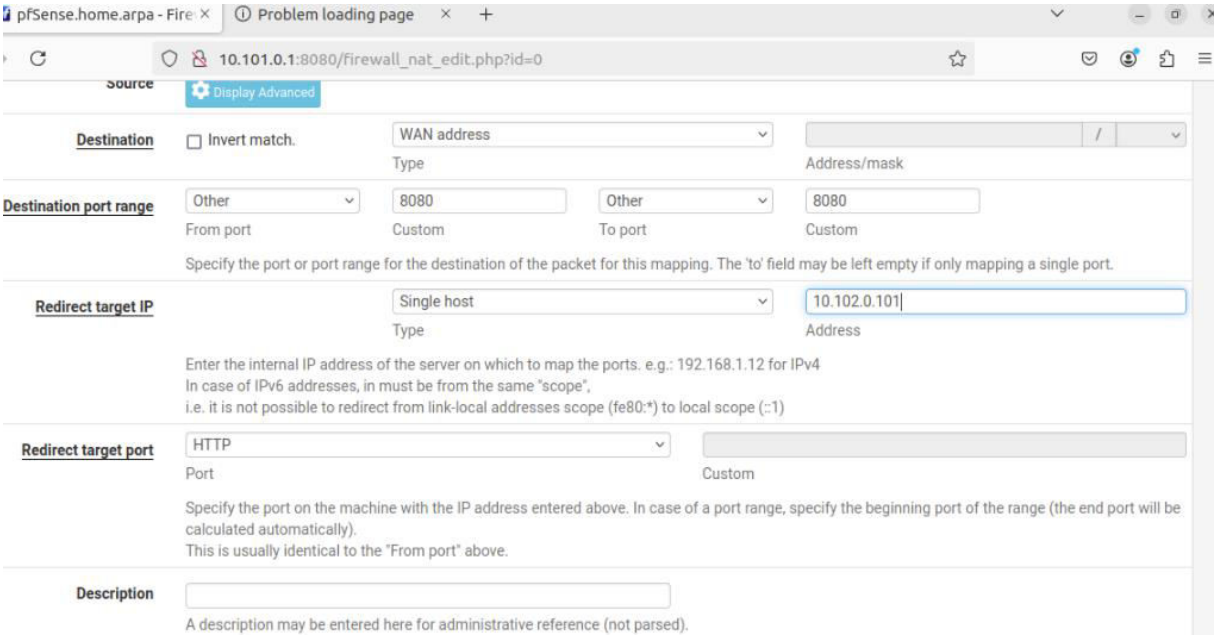




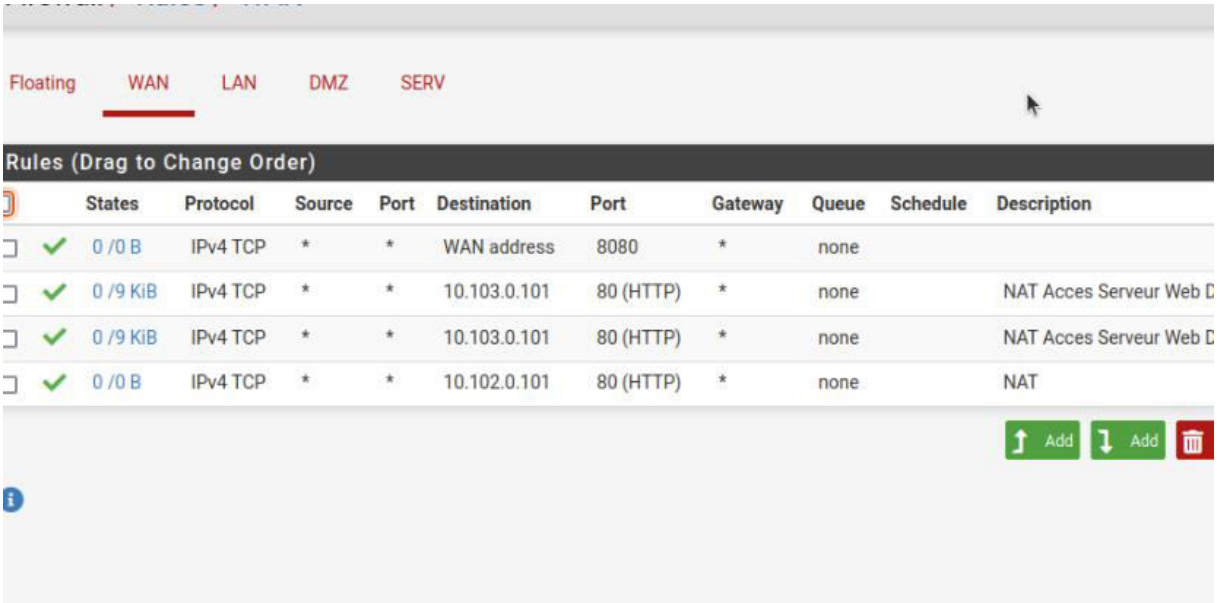
L'adresse IP semble avoir changée.

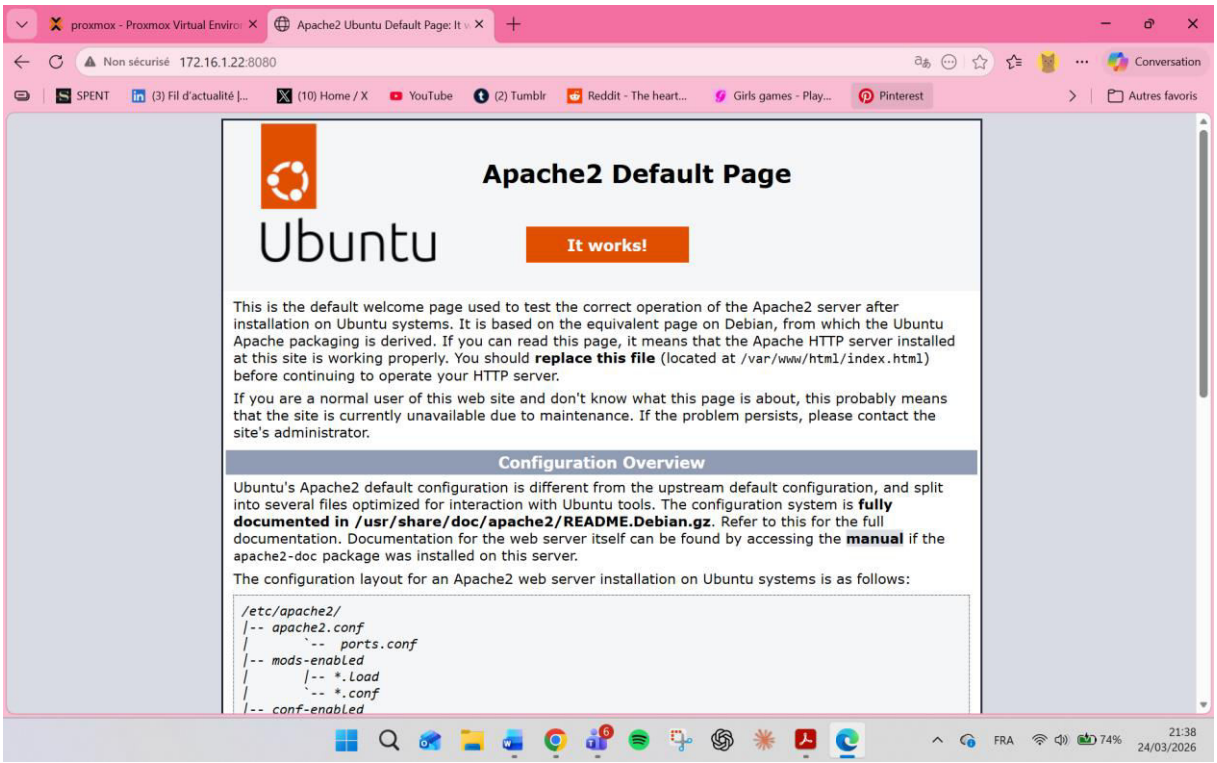
```
joyce@joyce:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether bc:24:11:15:da:ba brd ff:ff:ff:ff:ff:ff
    altname enp0s18
    inet 10.102.0.101/24 metric 100 brd 10.102.0.255 scope global dynamic ens18
       valid_lft 6924sec preferred_lft 6924sec
    inet6 fe80::be24:11ff:fe15:daba/64 scope link
       valid_lft forever preferred_lft forever
joyce@joyce:~$
```

Je modifie l'adresse IP de la règle



IP bien correctes





Tout est alors bien fonctionnel !